

Review

Recording Rainfall Intensity: Has an Optimum Method Been Found?

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Abstract: Many design principles for rain gauges that have the capacity to record rainfall intensity have been proposed or developed. These are here grouped into 15 categories, and the abilities and limitations of each are discussed. No standard or optimum method has emerged, despite more than 80 years of effort in the last two centuries, together with prior work from the 17th C onwards. Indeed, new methods continue to be explored for both point-based and area-wide collections of intensity data. Examples include the use of signal attenuation by rain along the tower-to-tower links of cellular phone networks, monitoring the speed of vehicle windscreen wipers, and exploiting the sound or vision from security and traffic-monitoring cameras. Many of these approaches have the potential to provide vastly more observation sites than conventional meteorological stations equipped with rain gauges. Some of these contemporary approaches seek to harness the potential of crowdsourced or citizen-science data. It is hoped that the present overview of methods will provide a guide for those wishing to collect or analyses rainfall intensity data for application in areas such as soil erosion processes, ecohydrology, agrochemical washoff, or urban flash flooding. Because rainfall intensity is one of the key aspects of the hydrologic cycle likely to respond as climate change and variability proceed, the choice of appropriate data collection methods has additional contemporary importance for the monitoring of regional and global precipitation changes.

Keywords: rainfall rate; rainfall intensity; rain gauge



Citation: Dunkerley, D. Recording Rainfall Intensity: Has an Optimum Method Been Found? *Water* **2023**, *15*, 3383. <https://doi.org/10.3390/w15193383>

Academic Editor: Stefano Luigi Gariano

Received: 9 August 2023

Revised: 23 September 2023

Accepted: 25 September 2023

Published: 27 September 2023



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1. Introduction

The intensity of rainfall varies from moment to moment. As Rodda & Dixon [1] observed, even recording ‘... the true rainfall at a point remains an elusive variable’. Intensity is a rain integral parameter, reflecting the combined arrival rate (water flux) of the range of drop diameters present in rain, falling at speeds that vary with diameter. At a single point on the ground, the intensity is actually zero in the moments between the arrival of successive drops and only becomes measurable as a (virtually) continuous variable when there are multiple drop arrivals across the substantial collecting area of a sensing device, such as the ~700 cm² area of a typical rain gauge collecting funnel. At the landscape scale, marked spatial variability in rainfall intensities and amounts arises over km scales (Fiener & Auerswald [2]) in part because wind and rain are steered by topography (Sharon [3], Hirose & Okada [4]), with small drops being more easily deflected by wind than larger drops. Though well-known (Blumen [5]), local-scale wind effects on rainfall have not been widely explored insofar as they affect intensity or storm pattern (intensity profile through the duration of a rainfall event) because rain gauge networks are rarely sufficiently dense to resolve these influences, especially in rugged or complex terrain where wind effects on rainfall might be most important. Gauges themselves of course perturb the local wind field near and above their collecting orifice, and this typically results in an under-catch as drops are swept past the gauge (Constantinescu et al. [6], Pollock et al. [7], Muchan & Dixon [8], Cauteruccio & Lanza [9]). The influence of topography further suggests that rainfall characteristics, including intensity, are likely to depend on wind direction,

and topographic steering will also influence the angle of rainfall arrival at the ground. Oblique rainfall can further affect rain gauge data (Crockford et al. [10]). The limited spatial and temporal resolution of satellite-derived rainfall products has been shown to result in biased data on rainfall intensities, over-estimating moderate and high intensity falls, and underestimating no- or low-intensity falls. This problem has been documented, especially in mountainous terrain where wind effects are complex (Yu et al. [11,12]).

Complicating the measurement of intensity are the rapidity of its fluctuations and the characteristic temporal intermittency of rainfall. Rain commonly stops and re-starts at intervals during a rainfall event (*intra-event* intermittency), and the temporary cessations may have durations extending from minutes to hours. If rain ceases for longer than a nominated interval, often 6 h, then when rain recommences, it is considered to mark the beginning of a new event. The rainless period is then classified as *inter-event* intermittency. Intermittency is linked to the measurement of rainfall intensity because if rainfall data are aggregated, for instance, to hourly totals, then it is likely that some raining time and some non-raining times are included, as well as rain of various different intensities. In the presence of intermittency, the actual intensity when raining would then be underestimated if the mean rainfall rate RR (mm h^{-1}) was calculated from $RR = P/T$, where P is the rainfall depth (mm) and T is the duration (h) through which P was tallied (15 min, hour, etc.). Consequently, time-aggregated rainfall data are generally unsuitable for the estimation of intensity, and data with high temporal resolution (seconds to no more than minutes) are needed.

Regardless of the various challenges to measurement just sketched, rainfall intensity and its momentary fluctuations exert an important influence on many landsurface processes, such as splash dislodgment and erosion of agricultural soils (Wu et al. [13]), triggering of erosive overland flow and mass movements (Guzzetti et al. [14]), and hazardous urban flash-flooding (Yang et al. [15]). Problems of urban drainage and related issues suggest that infrastructure design must allow for foreshadowed increases in short-term rainfall intensities (Sarhadi & Soulis [16]). Many studies have demonstrated that the necessary time resolution of rainfall data for successful modelling runoff from small urban catchments is no more than 3–5 min (Berne et al. [17], Lyu et al. [18]). Similar constraints apply to rainfall data to support soil erosion studies. In this field, intensity data alone are not sufficient; data on drop diameters as they influence splash dislodgment and other erosion processes are necessary. In many applications, such data are employed to calculate rainfall erosivity, which in turn is used to predict erosion rates. The primary rainfall data must be reliable to avoid error propagation through this sequence (Johannsen et al. [19]). There are also manifold influences of rainfall intensity on organisms and ecological processes, including the flight and feeding behaviour of birds and insects, the occurrence and spread of plant fungal diseases, and so forth. An interesting instance was provided by Araujo et al. [20]. They showed that rates of canopy disturbance in a tropical forest in Panama, resulting from branch- and tree-fall, were best accounted for not by wind speeds but by the incidence of 15 min rainfall rates above the 98.2th percentile, or approximately 24.3 mm h^{-1} . These disturbances and the creation of canopy gaps are key processes in tropical forest ecosystems. In forests and shrublands, and in many crops, rainfall intensity affects other processes such as rainfall partitioning within the canopy and the production of throughfall and stemflow, which may have ecological importance (Dezhban et al. [21], Zhang et al. [22]). Many other areas of significance within ecosystems could be listed. Furthermore, given that rainfall intensity is anticipated to undergo secular change, probably differentially in short and long rainfall events as climate change and variability proceed (Ban et al. [23], Kendon et al. [24], Li et al. [25], Armon et al. [26], Westra et al. [27]), intensity data are needed to validate model forecasts through the recording of actual secular changes in intensity. Change in extremes and their drivers is of considerable relevance (Hay & Williams [28]), not least because rainfall intensity has practical importance for society and the economy. Rainfall contributes to the erosion of wind turbine blades (Pryor et al. [29]) and may interfere with microwave communications. Indeed, the use of signal attenuation in mobile

phone networks and in satellite communications can be used to record rainfall intensity (see Christofilakis et al. [30] for a recent review). Indeed, differences in the scale and temporal resolution of different rainfall measurement methods (such as satellite-based data or ground-based rain gauges) pose significant technical challenges in validating observations. Among many complications is the fact that rain may only be falling along a part of the microwave link. Such difficulties are generally referred to as the “ground truth problem” (North et al. [31], Lebel & Amani [32], Yoo & Ha [33], Yoo et al. [34], Daly et al. [35], Ouyang et al. [36]): in other words, identifying just what the rainfall “truth” is is itself far from straightforward.

In addition to the ecological and societal areas already mentioned, reliable data on rainfall intensity is needed to guide rainfall simulation experiments, which are very widely applied to small-plot studies of agrochemical wash-off, soil detachment and transport, infiltration and overland flow production, and other important landscape processes (e.g., Shmilovitz et al. [37]). Such experiments often have durations ranging from a few minutes to an hour or so. In the absence of guiding data, many rainfall simulation experiments use intensities that are not designed to reproduce the possibly undocumented character of local rainfall but rather use arbitrary intensities, sometimes selected because of the kinds of pumps, nozzles, or other hardware that are available to the researchers, or else deliberately selected to generate results (e.g., by using a sufficiently high intensity that runoff actually occurred—see Dunkerley [38]). In most cases, the intensity of the simulated rain is held constant, with no intermittency and no fluctuations in intensity, in marked contrast to the character of natural rainfall (Dunkerley [39]). Wider availability of local rainfall intensity and intermittency data would support the improvement of rainfall simulation methods based on the local rainfall climate, enhancing the relevance of rainfall simulation studies to real-world problems in areas such as land management in agriculture, forestry, or mining.

Because there remain significant challenges to the instrumental measurement and recording of rainfall intensity and to validation and establishing the “ground truth”, the focus of this review is on the kinds of apparatus that have been developed. The design of rain gauges has a history spanning several centuries (Strangeways [40]), and developments are ongoing in both ground-based and remotely-sensed methods. Middleton & Spilhaus [41] and Habib et al. [42] provided overviews of the methods then available. The present review does not seek to be comprehensive, the relevant literature being very extensive indeed, but rather is intended to present some of the major approaches to measurement that have been devised and to highlight their use and limitations. Ground-based rain gauges have been, and are likely to remain for many years, the primary reference for rainfall measurement. Tapiador et al. [43] argued that this is partly because gauges supply direct recordings of a meaningful physical quantity, the amount of rainwater collected in a known interval of time, and the data thus have a clear interpretation and significance across many fields of application. In this sense, ordinary rain gauges continue to provide the “ground truth” against which indirect methods of measurement, such as microwave attenuation, can be judged and calibrated. Additionally, the established time series of rain gauge data collected at many thousands of observing stations globally, spanning many decades, would have great value if continued into the future using essentially the same kinds of well-tested and established apparatus. This would facilitate the analysis of secular changes in the rainfall climatology of the recording locations—including such aspects as the number of rain days; the mean rain day rainfall; the seasonal and diurnal occurrence of rainfall; the lengths of wet and dry spells; and importantly, the intensity of rainfall over land and the oceans.

Outline of the Remainder of the Paper

Most of the present review (Section 2) is devoted to a commentary on rain gauges, instruments whose purpose is the acquisition of rainfall data at a single location. The methods covered are as follows:

1. Historical sketches and obsolete approaches
2. tipping-bucket gauges

3. drop-forming and counting.
4. disdrometers
5. weighing gauges.
6. acoustic gauges.
7. optical, camera, and video-based methods
8. thermal gauges.
9. lysimeters
10. other electro-mechanical methods
11. nuclear methods.

In light of this diversity, it is pertinent to pose at the outset the question of whether an optimum method for recording rainfall intensity and intermittency has yet been developed or whether further development would still be of value. One tentative answer to this question is provided in the Conclusions to this paper.

In Section 3, attention is turned to those methods that provide wide-area rainfall intensity data, of the kind that is needed for many catchment hydrologic and runoff studies. The methods considered in this section are:

12. radar-based methods
13. rain measurements using microwave attenuation
14. seismic methods.

The treatment of these methods is less extensive than for Section 2, since all of the methods in Section 3 depend on some form of calibration or validation using point-based methods.

Finally, in Section 4, some miscellaneous and recent approaches are described as a single group:

15. miscellaneous (moving vehicles, windscreen wiper speeds, etc.)

Section 5 follows, and the paper finally presents some tentative Conclusions. We begin with a few remarks about the long history of rain gauge design.

2. Methods Suitable for Point-Based Observations of Intensity

2.1. Brief Commentary on Historical Developments

Ingenuity has led to the design of many forms of rain gauges—static catching gauges, non-catching gauges, recording gauges, and what were traditionally referred to as ‘rate of rainfall’ gauges. Some were simple devices with no moving parts, while others were elaborate electro-mechanical contrivances. A remarkable catalogue that illustrates the diversity of rain gauge designs from the early 1800s was presented by Kurtyka [44]. This report contains 193 Figures illustrating various rain gauges, together with a remarkable supporting list of 1079 references.

In considering intensity or ‘rate-of-rainfall’ gauges, the group most relevant for the present review, Kurtyka [44] adopted three categories:

- (a) tipping-bucket gauges (considered further below);
- (b) timed-entry gauges: in which a collector is exposed to rainfall for a fixed duration, e.g., employing multiple separate containers that are rotated at fixed time intervals to bring the next empty container under the rainfall collecting funnel. An example of this form of gauge, employing several chambers rotated by clockwork and each collecting rain for 1 min, was described by Sil [45]. Sequential rainfall samplers remain in use for studies of the changing composition of rainwater (e.g., Poissant & Béron [46]) but are no longer used for the measurement of rainfall amounts or intensity.
- (c) calibrated orifice type, involving interpretation of the rate of water flow through or over tubes, orifices, or weirs. Some of these gauges were drop-forming and counting devices. A gauge having no moving parts in which rain of increasing intensity progressively overtopped one of five small weirs, thereby diverting the amount of rainfall concerned into one of five separate collecting containers, was described by Scott [47]. Perhaps best-known among the calibrated orifice gauges is the Jardí gauge, produced in the 1920s. The Jardí gauge had a very large rainfall collecting funnel,

~1 m in diameter (e.g., see photo included as Figure 1 in Rossman & Wardle [48]), which led via a pipe to a chamber having a calibrated orifice and a carefully designed, hollow tapered float. According to Srivastava [49], the upper, larger float chamber (refer to Figure 1) was 40 mm in diameter, while the lower chamber was 20 mm in diameter. In increasingly intense rain, the float rose sufficiently high that the enlarged orifice between the upper and lower chambers that resulted from the float movement could allow the incident rain to drain, and a mechanism drove a pen to create a chart recording. Though notionally tracking intensity continuously and in a stepless way, testing has established that the response time of this instrument (i.e., the time required to adjust fully to a new rainfall intensity) was about 15 s (Cheng [50]). Nevertheless, its response to changing intensity was remarkably rapid. In a Jardí gauge with a very large collecting surface, significant evaporation of arriving droplets during breaks in rainfall would be a possibility. Chen [51] noted problems of solid debris entering the float chamber of the Jardí gauge and of the need for regular cleaning and lubrication of the mechanical linkages to reduce frictional drag. Difficulties with accurately reading the chart recordings have constrained the analysis of Jardí data, according to Llasat & Puigcerver [52], though they were able to digitise historical charts and resolve intensities from them. Gauges of this kind appear no longer to be in common use. Nevertheless, the Jardí gauge is an example of an instrument designed specifically to enable rainfall intensity to be recorded.

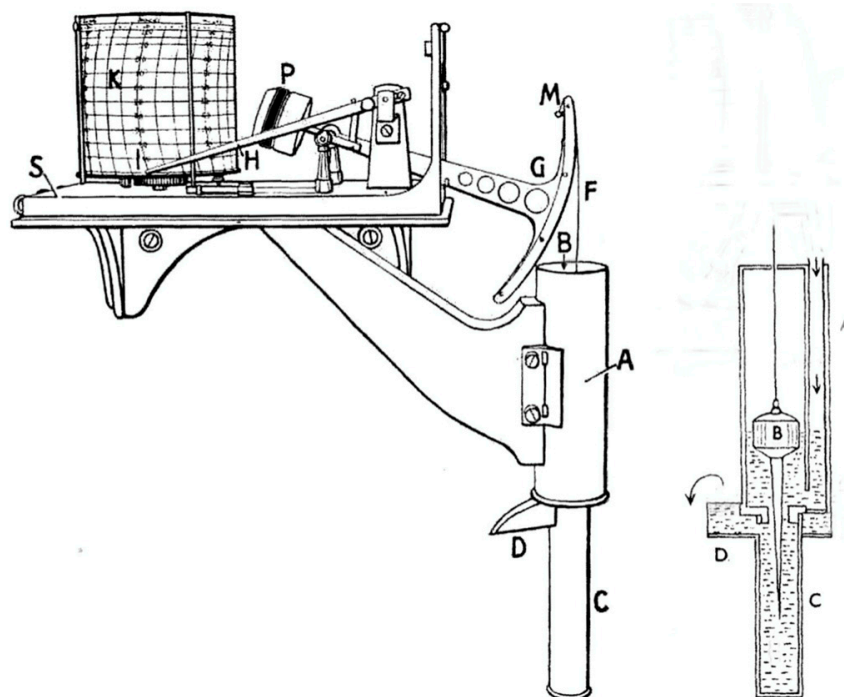


Figure 1. Schematic of a Jardí rate-of-rainfall gauge. The tapered float (B), shown in cut-away view at right, rises and falls with intensity, changing the dimensions of the orifice beneath it. Source: https://en.wikipedia.org/wiki/Ramón_Jardí_i_Borrás (accessed on 2 April 2023). Reproduced in accordance with license CC-BY-SA-3.0.

Published data collected using Jardí gauges often show very high intensities ($>250 \text{ mm h}^{-1}$, e.g., see Figure 7 of Chen [51]). Indeed, Llasat & Puigcerver [52] reported that the highest rainfall rate ever recorded in Barcelona, extracted from digitised historical Jardí chart records, was a remarkable 586.8 mm h^{-1} . It is not clear how successful the Jardí gauge was in tracking intensity fluctuations in rain of low intensity, which remains challenging to this day. Nevertheless, the ability to establish the existence of very high intensities is one of the important contributions of intensity gauges such as the Jardí design. These high intensities,

which may last for periods of only minutes, cannot be revealed in rain data that is reported for fixed collection times, such as every 15 min, or every hour.

2.2. Tipping Bucket Gauges and Related Devices

Tipping-bucket rain gauges (TBRGs) have become the most common type of recording rain gauge, and the see-saw mechanism is used to record rainfall amounts in increments that are often 0.2 mm and less often 0.1 mm or 0.5 mm, set by the capacity of the buckets. The time of each tip event may be individually logged, or the running total number of tips may be logged less frequently. TBRGs are subject to many of the same sources of error in rainfall amounts as other types of gauges, including exposure and wind under-catch effects, evaporative losses, dynamically changing calibration, and the under-recording of intense rain (Segovia-Cardozo et al. [53]). They are widely used to estimate rainfall rates and intensities, though their design is not particularly suitable for recording intensity or for estimating rain rates over short integration times (e.g., less than tens of minutes). The key problem is that there may be an intensity-dependent time delay before bucket filling and tipping actually records the rain, and a partly-filled bucket may fail to tip when rain ends, such that the last recorded tip might be recorded some time before the true cessation of rainfall. Consequently, the actual duration of rainfall is not well resolved, and the extent of the uncertainty of the duration itself depends on the intensity. If the true duration of rainfall is not known, then even if the total amount of rainfall recorded in, say, 1 h, is accurately known, the intensity cannot be estimated with confidence. This issue makes the TBRG less than ideal as a source of “ground truth” intensity data, particularly at locations where rainfall is commonly intermittent or of low intensity.

TBRG mechanisms are subject to systematic errors at high rainfall rates. These arise from the rain that arrives from the collecting funnel during the finite time required for the bucket mechanism to tip. Some of this water is allocated to the already-full bucket as it tips, resulting in an underestimation of the amount of rainfall. There are many studies in which this bias has been quantified and which recommend dynamic calibration such that the bucket capacity is not regarded as fixed, for instance, at 0.2 mm, but rather the effective capacity changes with the intensity. Examples include Duchon & Biddle [54], Shedekar et al. [55], Sypka [56], and Liao et al. [57]. The magnitude of the bias depends on the bucket and raingauge design; however, in laboratory tests, Sypka [56] found that the bias exceeded 10% at 100 mm h^{-1} , and became even larger at higher intensities. There may also be impulsive effects from large step-changes in rainfall intensity. Many TBRGs include a small syphon, whose purpose is to deliver a steadier flow of water to the tipping buckets. However, the various designs for the syphon also influence the performance of a TBRG (Choi et al. [58]).

An important modification of the TBRG involves the weighing of the entire bucket mechanism as it fills, in order to eliminate the loss of intensity information during filling time. Lee [59] and Kim & Lee [60] proposed and tested such a device in both the laboratory and the field. Their weighing TBRG design had a resolution of 0.01 mm of rainfall. This approach is currently embodied in commercial rain gauges such as the Lambrecht “Rain[e]” (<https://www.lambrecht.net/>, accessed on 2 April 2023) that combine a tipping bucket mechanism with a weighing system. The Lambrecht device has a collecting area of 200 cm^2 (considerably smaller than a standard rain gauge), and the stated resolution of intensity measurements is 0.001 mm h^{-1} . The “Rain[e]” is reported to be capable of working at intensities of up to 1200 mm h^{-1} .

Recently, Hu et al. [61] have described the development and testing of a modified form of TBRG in which the progressive filling of a single bucket lifts a float whose motion generates electric power tribo-electrically. Choi et al. [58] evaluated a modified form of TBRG that was equipped with two bucket mechanisms: an upper system with a sensitivity of 0.1 mm and a lower mechanism with a 0.5 mm sensitivity. The lower bucket was designed to catch any splash from the upper bucket, and in data processing, the rainfall in

an interval was taken as the larger of the two rainfall depths recorded by buckets 1 (upper) and 2 (lower).

There are more complex variations on the TBRG principle of measuring small, successive quantities of rainfall and recording the time of arrival of those small amounts. Mink & Forrest [62] developed a rotating gauge with 12 rainfall collecting chambers bored into a stainless steel disc that could be rotated to bring a fresh chamber under the collecting funnel (Figure 2). The volume captured in each chamber as it rotated was 0.1 mL, which, with a collecting area of 200 cm², yielded a notional sensitivity of 0.005 mm, or 40 times finer resolution than a standard 0.2 mm TBRG. Drabbe [63] subsequently described a device that had two chambers, one being filled while the other was emptied, with the switching controlled by electrical solenoid valves. The device provided a measurement resolution of 0.1 mm. Both devices were quite complex, required very tight manufacturing tolerances, and required regular greasing and other maintenance of moving parts. A further gauge mechanism was described by Onacak & Yurur [64], again employing electronic solenoid valves to control the entry and drainage of water from a measuring system. Tabada & Loretero [65] proposed a solid-state electronic alternative to the use of reed switches, activated by moving magnets, to reliably detect bucket tip events. While offering high precision and greater sensitivity, none of these complex electro-mechanical alternatives to the conventional TBRG appears to have found application in the routine recording of rainfall intensity. In any case, being based on a collecting funnel to gather the rainwater for measurement, all of these designs become subject to potential evaporative losses and to time-lags arising from the delay before the funnel becomes sufficiently wet to allow water to drain into the measuring system. Further, the reliance on a collecting funnel introduces potentially problematic interference with airflow in the vicinity of the gauge. These potentially further detract from the high sensitivity of any mechanism that relies on a large collecting funnel. In other words, the devices mentioned are potentially good at recording the volume of water delivered by the collecting funnel. However, as is often the case, the question left unanswered is to what extent this can confidently be accepted as reflecting the intensity of open-field rainfall in the immediate vicinity of the gauge.

There are several approaches to the estimation of rainfall intensity from conventional TBRGs. The most direct is to analyse the time that elapses between successive bucket tips and assume that in the relatively brief time that often elapses between them, the rainfall intensity is constant. On that basis, the rainfall rate RR can be estimated from moment-to-moment from the relation.

$$RR = V / (T_2 - T_1) \quad (1)$$

where V is the bucket capacity in mm of rainfall (typically 0.2 mm) and T₁ and T₂ are the times of two successive bucket tip events (h). This approach was adopted by Williams & Erdman [66] and Costello & Williams [67], among others. There is no wholly acceptable way to extract 1 min intensities from TBRG data, since the tip events are not synchronised to clock time (Stagnaro et al. [68]). There are approaches that seek to derive intensities during clock minutes (also called “rain minutes”), defined as clock minutes during which a tip or tips occur. The rainfall represented by those tips is simply allocated to the clock minute in which they occurred, but of course this need not be the case. Such approaches are generally subject to significant error, especially in low- to moderate-intensity rainfall. Wang et al. [69] explored these issues and the possible benefits of post-processing the estimated 1 min data by fitting cubic spline curves. However, no post-processing method can eliminate the effects of the quantization set by the bucket filling time, and 1 min rainfall data derived from TBRG measurements must be assessed for reliability on a case-by-case basis.

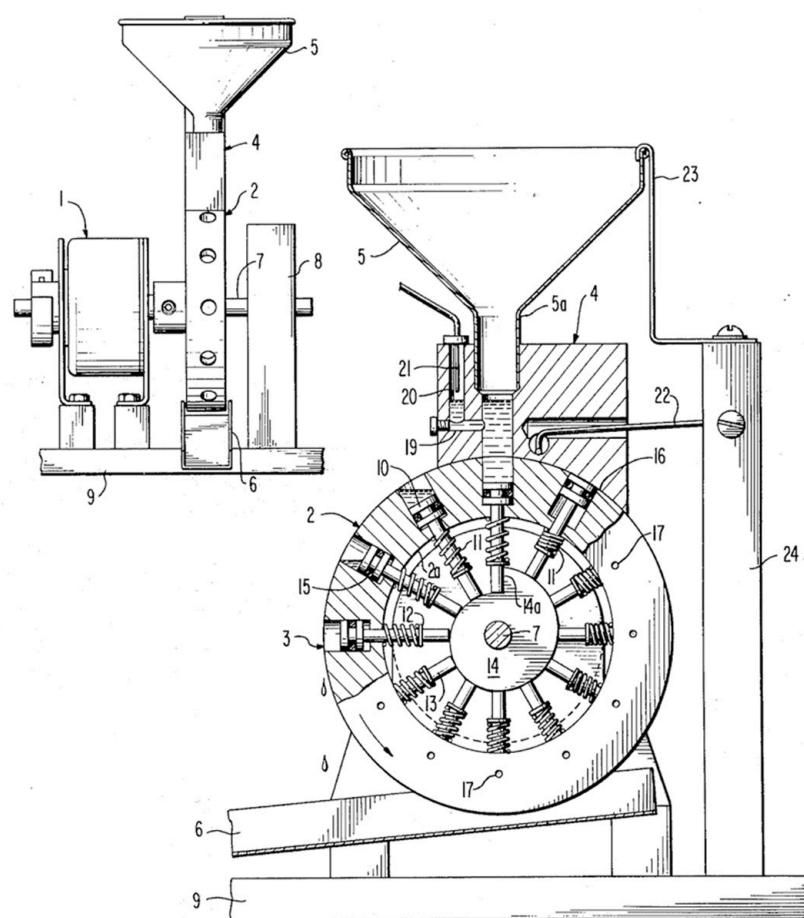


Figure 2. Schematic diagram of the Mink & Forrest rain gauge. Source: United States Patent No. 3958457 (inventor: J.W. Mink, May 1976). The part numbers are explained in the US Patent document.

It is worth mentioning here the use of syphon-based gauges as an alternative to TBRG designs. In these gauges, self-emptying occurs much less frequently than in a TBRG. An early form of syphoning gauge was the Dines tilting-syphon gauge (designed by British meteorologist William Henry Dines; see Strangeways [70]). This was a catching gauge in which the reservoir holding the water contained a float that rose as the container filled progressively. When the container was full, the rising float released a catch, which allowed the unbalanced container to tip and empty, following which it returned to its original position. Later syphoning gauges used a self-siphoning mechanism with no moving parts (such a device is manufactured by the R.M. Williams Company, Wilmington, DE, USA) (<https://www.youngusa.com/>, accessed on 2 April 2023). In this gauge, the syphon action is triggered when 50 mm of rain has been received. When full, these gauges self-siphon to empty the water storage vessel, a process taking about 30 s, during which rainfall is not correctly tallied. As for many catching-type gauges, the collecting area is relatively small (100 cm^2), presumably to limit the volume of rainwater collected; however, this reduces their appropriateness for the recording of rainfall rate owing to the likely under-catch of large drops. Self-siphoning rain gauges have been widely deployed for data collection at sea (Serra et al. [71]). These gauges can record accumulated rainfall well but suffer from the same source of error as TBRGs, viz., their failure to record the rainfall that arrived while the collecting vessel was tilted and emptying (or siphoning). A modified syphon gauge, designed to eliminate some of the errors of the Dines gauge, was later proposed by Nothmann [72] but appears not to be in contemporary use. The emptying time needed for the Nothmann gauge was reported to be only 5–7 s, which would nevertheless result in significant errors but for the adoption of a weighing mechanism separate from the self-emptying system. There are additional commercially available self-emptying rain gauges,

such as the ‘RainBal’ device (<https://www.hyquestolutions.eu/>, accessed on 2 April 2023), that likewise employ weighing mechanisms and a tipping, self-emptying bucket.

2.3. Drop Forming and Counting Gauges

In seeking a rapid response to intensity fluctuations, gauges that form and count a sequence of drops were developed. The key advantage of drop-counting gauges is that they eliminate the time delay associated with the filling of a TBRG bucket. In an early form of drop-counting gauge, Bibby [73] described the ‘rainfall chronograph’ used at the UK Kew Observatory. Developed by Whipple, this device used a funnel to collect rainwater, which then apparently dripped onto a small, hinged paddle, which tipped and momentarily closed an electrical contact. Adkins [74] described a drop-forming gauge in which the drops fell onto a wire grid, thus momentarily closing a circuit. In all such gauges, the drop-forming systems need to be carefully designed so as to produce drops of constant volume. In many subsequent gauge designs, a standard collecting funnel delivers water to a capillary tube of about 3 mm inside diameter, whose tip releases the series of free-falling drops. In the case of the drop-counting gauge described by Norbury & White [75], the drop volume was ~ 0.07 mL. The drops were then counted optically as they fell through an infrared beam. Given the small volume of a single water drop, much faster response is possible than with a conventional TBRG, and a single drop is equivalent to a sensitivity of ~ 0.006 mm, or more than 30 times that of a 0.2 mm TBRG. Instruments of this kind, typically with a 10–15 s response time, have been described by Norbury & White [75], Sharma et al. [76], and Hosking et al. [77]. Figure 2 in Norbury & White [75] presents the high-resolution graph of about 50 min of rainfall from a thunderstorm, the highest intensity being almost 140 mm h^{-1} .

Stow & Dirks [78] and Stow et al. [79] presented further refinements of drop-forming and counting gauges, including a high-performance drop-forming system. Drop detection was not optical, but rather, each falling drop momentarily contacted two fine electrodes, thus briefly closing a circuit. They adopted a 145-mm-diameter collecting funnel (with a 30° side slope), and their drop-forming system delivered drops of a nominal 3.0 mm diameter. The resolution was thus ~ 0.006 mm of rainfall depth. Figure 6 in Stow & Dirks [78] shows the data collected during a short, intense rainfall event, with the data having been tallied at three different resolutions: 15 s, 1 min, and 6 min. The apparent reduction in intensity and the altered apparent shape of the hyetograph in that figure demonstrate the need for a resolution of just seconds if intensity bursts are to be resolved and recorded.

Sansom & Gray [80] provided experimental data on the performance of a drop-counting gauge (referred to as the ‘RIG’, for ‘rain intensity gauge’), which was based on the design of Stow & Dirks [78]. They employed a co-located TBRG and a Stow gauge as reference devices. Sansom & Gray [80] reported that their gauge could record the occurrence of rain at 0.1 mm h^{-1} with a lag of only 1–2 min, and that the problematic trickle phenomenon (when rain is so intense that the water emerges from the capillary tube as a trickle rather than as separate drops) only arose in rain at $>100 \text{ mm h}^{-1}$. Overall, the temporal resolution (funnel draining time plus drop formation and release time) was ~ 6 s. Various issues were referred to in this study, including the need to ‘age’ newly-made devices until their calibration stabilized. This is observed for many kinds of gauges and relates to the accumulation of deposits from the water that may be hydrophilic, including on the drop-forming tube.

As for conventional TBRGs, dynamic calibration is needed because the drop volume is not in fact constant across all rain rates, and indeed, an upper limit is imposed at high rainfall intensities when the stream of drops becomes a continuous trickle (Stagnaro et al. [81]). Pickering et al. [82] reported such problems at intensities of $>50 \text{ mm h}^{-1}$, which is by no means the most intense rainfall likely to be encountered at many locations. Maintenance is also an issue, particularly from the effects of detritus entering the capillary tube or drop-forming system. According to Stagnaro et al. [81], drop-counting gauges tend to overestimate rainfall intensity in the low- to intermediate-range.

Applications of drop-counting gauges include Sichoix & Benoit [83] who employed a network of 10 ‘Pluvimate’ drop-counting gauges in a study of orographic rainfall in Tahiti. These devices count the drops via an impact transducer. See <https://www.driptych.com/pluvimate.html> (accessed on 2 April 2023) for examples of hyetographs and operational details of this commercial instrument. Sarkar et al. [84] also employed drop-counting gauges with 10 s resolution to provide intensity data for a study of the attenuation of microwave signals.

2.4. Disdrometers—Optical and Electro-Mechanical Drop Sizing, Counting, and Fall Speed Measurement

Instruments capable of measuring the distribution of drop sizes in rain—‘disdrometers’ (for ‘distribution of drop diameters’)—have been designed using a wide range of design principles. These include the brief occlusion of a light beam or light sheet as a drop passes through it, the extent and duration of which can provide information on both the drop diameter and its fall speed. Essentially, they all offer rapid measurement capacity, and often drop counts are tallied for 1 min and the resulting data recorded. The drop-detection technologies employed in disdrometers include optical devices, including lasers, mechanical impact devices, and radar (considered separately in Section 2.9 below).

Disdrometers were originally dedicated drop-sizing instruments, designed to explore the drop size distributions of various kinds of rainfall and the change in drop size distribution (DSD) with rainfall intensity. However, drop size information is insufficient to derive rainfall intensity, for which the fall speed is also needed so that the flux can be determined. In some cases, primarily with impact disdrometers, fall speeds were not measured but rather estimated from recorded drop sizes by using pre-established empirical relationships based on lab experiments and linking drop size and the terminal fall velocity.

As outlined by Sheppard & Joe [85], many disdrometers that in principle have the capability of recording intensities with high temporal resolution (such as impact disdrometers and some optical drop measuring sensors; see below) do not measure intensity directly. Rather, they determine parameters such as drop counts, diameters, and fall speeds. From these, the rain aggregate parameters such as depth and rain rate are estimated by summing across all diameter classes that can be sensed by the particular device. This poses the need for validation by some independent method, which Sheppard & Joe [85] argued is still lacking. Other problems that affect some of these designs include the rather small disdrometer sampling orifices, often of about 50 cm², which are deliberately used to reduce the chance of two (or more) drops arriving simultaneously and being counted as one large drop. Furthermore, as Sheppard & Joe [85] point out, there is a lack of internationally validated and accepted reference measurement methods able to generate “ground truth” against which the accuracy and reliability of any new method or disdrometer instrument can be assessed. Some disdrometers, for instance, are calibrated by dropping metal or glass spheres of a few accurately known sizes into the detecting mechanism. Some of these issues will be explored further below.

Among the most well-established impact-sensing devices, the Joss-Waldvogel disdrometer (JWD) has gained wide acceptance. It uses a low-mass target that is exposed to rain and whose movement generates a voltage via the motion of a magnet in a sensing coil. The current model RD-80 has become a standard for the measurement of drop size distributions (<https://distromet.com>, accessed on 2 April 2023). The device consists of a conical Styrofoam cone capped by a replaceable smooth membrane that is exposed to rain. Movement of the Styrofoam target (which must exceed the motion resulting from any local acoustic noise) displaces a magnet in relation to two sensing coils, one of which detects the small induced voltage spike, and the second generates a force that resists the motion of the cone, such that there is little actual displacement or resonance. The drop-collecting area is 50 cm², and data are allocated to 127 drop-diameter channels. Drops smaller than about 0.3 mm are not detected. Taking the collecting area of a typical conventional rain gauge to be ~700 cm², it is apparent that the collecting area of the JWD is <10% of that size, so that in

fact it would require multiple JWDs to offer the same collecting area. The small sensor size is both an advantage by tending to reduce the occurrence of multiple simultaneous drop impacts that might be tallied incorrectly and a disadvantage by reducing the likelihood that rarer large drops will actually be counted in proportion to their true volumetric abundance in the air above the device. Given that the JWD only records drop sizes, it does not provide a direct measurement of rainfall intensity.

Other approaches to recording drop arrivals by measuring drop impacts have been proposed, including the device explored by Lu et al. [86] which was found to be capable of recording drops in the range of 0.4–2.0 mm in diameter and up to 50 drop arrivals per second over the 28.3 cm² sensing area of the device. Madden et al. [87] developed an impact sensor using a ceramic disc of 50 mm diameter. This was used to record the rainfall power (in mJ cm^{−2} min^{−1}) which was found to correlate quite well with the rainfall intensity as recorded by a 0.1 mm TBRG. Antonini et al. [88] presented a design for an impact rainfall sensor based on a low-cost piezoelectric transducer.

Because the JWD does not sense the fall speed of the drops, the rain rate (and other rain integral parameters) cannot be determined directly. As a result, the fall speed for the various drop diameter classes has to be adopted from independent work on fall speeds in relation to diameter, such as that of Gunn & Kinzer [89]. Hence the precision of intensity data derived from a JWD is influenced by the particular diameter-speed relation that is adopted in data reduction. Additionally, fall speeds are only well-known in still air with vertical fall trajectories. In strong and gusty wind, there is no certainty that empirical diameter-fall speed relationships would still hold. Indeed, the behaviour of fall speeds in the range of environments where rainfall observations are required remains incompletely known, and super-terminal and sub-terminal speeds have been documented (Montero-Martinez et al. [90], Niu et al. [91], Brangi et al. [92], Chatterjee et al. [93]).

In contrast to electro-mechanical impact disdrometers, many optical disdrometers (and radar-based systems, see below) are able to determine both the diameter (by shadow dimensions) and fall speed (from successive images closely spaced in time or by two closely spaced light beams or sheets) of incident drops. Examples include the optical disdrometer design by Löffler-Mang & Joss [94], the Ott Parsivel (<https://www.ott.com/>, accessed on 2 April 2023) (for ‘particle size and velocity’), various video disdrometers, HVSD (Hydrometeor Velocity and Shape Detector, Barthazy et al. [95]), the video precipitation sensor (Liu et al. [96]) and the HHI (Holographic Hydrometeor Imager, Kaikkonen & Makynen [97]), the Thies disdrometer, as well as many others. The commercial Thies (<https://www.thiesclima.com/>, accessed on 2 April 2023) laser disdrometer was found by Fehlmann et al. [98] to consistently underestimate rain rates by 16.5%. Typically, these devices have either a sensing area, if drops fall through a laser light sheet, or a sensing volume, if drops fall through a collimated beam of light. For instance, the video precipitation sensor (Liu et al. [96]) has a sensing volume of 360 cm³ because drops pass through a cylindrical light beam. The holographic system (Kaikkonen & Mäkynen [97]) has a measuring volume of 670 cm³. Video disdrometers that use light sheets typically have sensing areas of ~100 cm² (e.g., Salles et al. [99], Schuur et al. [100]) and hence also have only ~15% of the collecting area of a standard meteorological rain gauge. Drop sizes across a wider range than offered by the JWD can be measured by many optical disdrometers with quite high precision.

Given the precision with which incident drops can be measured, does the resulting DSD accurately represent the DSD in a volume of air above the sensor? The answer will be in part dependent on the relatively brief sampling time (typically 1 min). On the assumption that the exponential Marshall-Palmer DSD was applicable, Dunkerley [101] showed that the arrival rate of large drops can become too small to be reliably tallied by a disdrometer with a small sensing aperture and a short tallying time. Tapiador et al. [102] found that an optical disdrometer could underestimate rainfall intensity by up to 70% as a consequence of the under sampling of large and less frequent drops. The correct tallying of large drops is important if intensity is to be correctly recorded, since one 5 mm drop delivers the same

volume as ~126 drops of 1 mm diameter. This limitation is reduced but not eliminated in video disdrometers with a somewhat larger sensing area.

Disdrometers often also undercount small drops. The implication of the underrepresentation of small or larger drops is that the rainfall aggregate parameter and rainfall rate will be underestimated. This has, in fact, often been found. Feloni et al. [103] found an underestimate of 2% in JWD data. In comparing the Parsivel optical disdrometer with TBRG data, both aggregated to hourly data, the disdrometer was underestimated by 30%. This was attributed to the large number of small drops in the local rainfall (Cherrapunji, India), which were not tallied correctly by the disdrometer. Clearly, the local DSD characteristics of the rainfall influence disdrometer performance. Jaffrain & Berne [104] found that a Parsivel only underestimated the aggregate rainfall of a 15-month period by 4.3%, judged against a TBRG; however, such results cannot confidently be taken to apply to other locations.

Disdrometer data can be perturbed by wind. Drops entering the sensing area of a device such as a Parsivel disdrometer obliquely take a longer path through the light sheet, and this results in a slower estimated fall speed. Friedrich et al. [105] showed that the oblique entry of raindrops into the wind resulted in Parsivel disdrometers reporting anomalously large and slow drops. In attempting to reduce the resulting bias, they demonstrated the utility of an articulating disdrometer, which was mechanically moved so as to maintain the orientation of the sampling window orthogonal to the direction of drop arrival. Wind may affect drop shape asymmetry and hence fall speed. Thurai et al. [106] showed that gusts could reduce fall speeds by 25–30%. Lin et al. [107] showed that for wind speeds $> 14 \text{ m s}^{-1}$, the error in 5 min rainfall recorded by a Parsivel disdrometer was a 22.4% underestimation compared to a TBRG. Biases in Parsivel data were also documented by Tokay et al. [108] in a study of two Parsivel devices, a JWD, and two TBRGs. The JWD also recorded a significant divergence in results compared with the TBRGs. Wind effects were shown by Pang & Graßl [109] to result in heterogeneous rainfall over a few seconds or a few meters. Wind effects were also highlighted by Bezak et al. [110] and Capozzi et al. [111], and Chinchella et al. [112] in a study of the Thies laser disdrometer. Evidently, although optical disdrometers can measure drop sizes reliably, if drop arrival is strictly vertical, they are prone to significant errors under windy conditions.

A common way to evaluate disdrometer data are to make comparisons with their aggregated rainfall totals at hourly aggregation time, or over some months or a year, with data from other gauges such as a TBRG. Such studies (e.g., Islam [113]) have indicated discrepancies of 10–20%, which seems like a reasonable general level of uncertainty to attach to such data. Islam [113] compared aggregate rainfalls from a JWD, a TBRG, and several rapid-response (drop counting) gauges. In the paper, it is accepted that the JWD data are ‘correct’ and that divergence in the data of other gauges represents error. Nevertheless, Tokay et al. [114] reported significant differences in the data collected by six collocated JWDs, and Tapiador et al. [102] reported significant discrepancies among 14 co-located laser disdrometers. Further, in comparing DSDs from four types of disdrometers (2D video, X-band radar, micro rain radar, and JWD), Chang et al. [115] identified the JWD as the least accurate, attributing this to the small volume of air sampled by this device. In a study carried out at Cherrapunji, India, using a Parsivel disdrometer, it was found that, compared to a TBRG, the disdrometer underestimated hourly rainfalls by 30%. Underestimation by Parsivel disdrometers was also reported by Jaffrain & Berne [104] and by Wen et al. [116]. In some cases the bias has been shown to result primarily from the limited capacity of disdrometers to record the arrival of small drops (Park et al. [117], Wen et al. [116]).

Owing to the different results obtained by co-located devices of the same type, it has proven useful to have multiple identical, collocated devices. The differences in the data produced by each can provide an estimate of the uncertainty that is likely to be associated with reliance on a single measuring instrument. Annella et al. [118] demonstrated the application of a triple-co-located installation consisting of two disdrometers and a rain gauge.

2.5. Weighing Rain Gauges

The use of progressive weight gain in a rain receptacle is an appealing approach to the recording of rainfall intensity and has been adopted in various commercial rain gauge designs such as the Ott 'Pluvio²' (<https://www.ott.com/>, accessed on 2 April 2023) and the 'Geonor' gauge (<https://geonor.com/>, accessed on 2 April 2023). The use of weighing in modified TBRGs was noted above; however, the gauges considered here rely on weighing alone.

Hanson et al. [119] presented the design and tests of a load-cell-based weighing gauge with a voltage output proportional to the weight of contained water. This device recorded the increasing weight as frequently as 1 min intervals, provided at least 0.25 mm of rain had fallen during that minute (that is, at a minimum rate of 12 mm h⁻¹, which is moderately intense rain). The device was intended to record the start and end times of rainfall with good precision. Weight measurements were always recorded at 15 min intervals, synchronised to clock time. These gauges had a capacity of 300 mm of rain with an 8 inch (20.3 cm) diameter collecting orifice.

Both the Ott and Geonor commercial gauges have buckets that store the collected water that is weighed; these buckets have to be emptied manually when (or before) their capacity is reached. The Ott gauge uses a load cell to record weight, while the Geonor gauge uses three 'vibrating wire' sensors, whose resonant frequency depends upon the load of collected rainwater held in a bucket that is suspended from them. The Geonor gauge has a collecting area of 200 cm², and the sensitivity declines as the storage capacity (weight) of the collecting bucket area is increased. For example, for a Geonor gauge having a capacity for 600 mm of precipitation, the sensitivity is 0.05 mm; however, this declines to 0.1 mm for a 1500 mm gauge. Both types of weighing gauges have theoretically very good precision for the depth of rainfall received; however, their sensitivity in terms of intensity is not equally good. The Ott gauge is available with 200 cm² and 400 cm² collecting areas, and according to the manufacturer's technical data, record intensities ≥ 12 mm h⁻¹. Based on lab tests in which water was fed into an Ott gauge by using a peristaltic pump, Saha et al. [120] suggest a lower intensity limit of 7 mm h⁻¹ when acquiring 1 min of data. Given that the Ott (and Geonor) gauges store the rainwater for weighing, their capacity is limited. The 200 mm orifice Ott gauge can record a maximum of 1500 mm of rain; however, the 400 mm orifice version is limited to 750 mm of rain. The Geonor gauge has similar maximum storage depths. Given their operation and capacity for long-term deployment, these gauges require oil to be added so as to reduce evaporative losses (and also, in cold conditions, antifreeze).

Various sources of error can arise with weighing gauges, especially in very intense rainfall. Bodtmann & Ruthroff [121] proposed correction procedures designed to permit 1 min rainfall rates to be estimated reliably. Temperature fluctuations can result in spurious rainfall being recorded (Knecht et al. [122]). Devine & Mekis [123] reported undercatch in a study of the Geonor weighing gauge. Approaches to reducing noise in the data from weighing gauges were explored by Nayak et al. [124]; Ross et al. [125] explored four approaches to the post-processing of weighing gauge data to reduce the effects of mechanical and electrical interference, as well as evaporative loss.

2.6. Acoustic Intensity Gauges

The sound of raindrops striking a detector of some kind has been exploited to record rainfall arrivals and allow rainfall intensity to be determined. The acoustic energy released by rain striking a metal roof has been a subject of considerable study in the acoustic design of buildings (Schmid et al. [126]), and it is known from that work that the sound energy is linearly related to the logarithm of rainfall intensity. Figure 3 presents an example from Dubout [127].

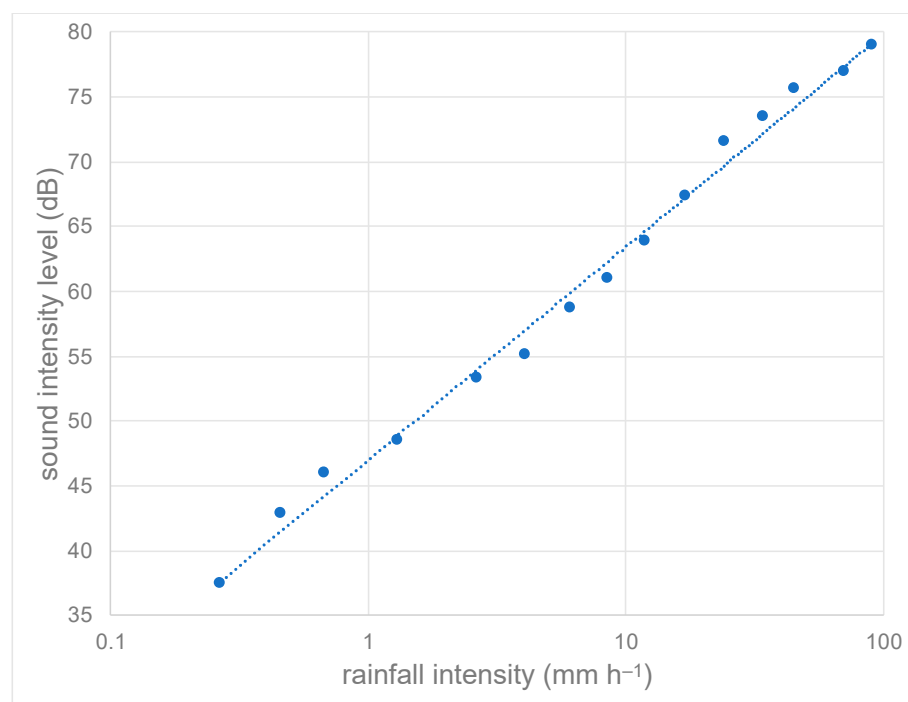


Figure 3. The relationship between sound intensity (in dB) and rainfall intensity (mm h⁻¹) was established by Dubout (1969) [127] in a small experimental shed roofed with galvanised metal sheets. The rainfall intensity data were derived from a TBRG located nearby. The acoustic signal and TBRG tip events were recorded on a chart. The plotted points are the averages of the data collected during 1 min sampling times and plotted in intensity classes. The regression model (dotted line) linking dB and intensity (I) has the equation $\text{dB} = 7.13 \ln(I) + 47.0$ ($r^2 = 0.99$).

Thus, for a given calibrated building roof structure, it is in principle possible to use the recorded sound pressure measured beneath the roof as a record of the outside rainfall intensity integrated over the roof area. Given that acoustic sound recordings are frequently made at a sample rate of 44.1 kHz, an audio recording has ample time resolution to resolve the signal of individual drops striking a sound source if the collecting area is not so large that there are multiple simultaneous drop arrivals. In some applications, sophisticated damping of the resonant membrane or other sensor is used, together with numerical signal processing, in order to yield a system capable of discriminating individual drops according to their diameter by interpreting the acoustic signal of each drop impact to form an acoustic disdrometer.

The acoustic recording of rainfall has been widely adopted for use at sea, generally with submerged sensors that detect the sound arising from drop impacts on the water surface and the bubble bursting associated with this (Black et al. [128], Ma & Nystuen [129,130], Nystuen [131]). The nature and origin of the sound produced by drops striking a water surface were explored by Prokhorov & Chashechkin [132]. To date, acoustic recording of rainfall has been adopted in land-based research in only a small number of experimental studies (Trono et al. [133], Guico et al. [134], Avanzato & Beritelli [135], Dunkerley [136]), and has not yet found application in long-term rainfall monitoring.

One important advantage of acoustic methods for recording rainfall intensity is that no collecting funnel is needed since drops fall directly onto the sensor. This eliminates the time lags that arise when a large funnel needs to be wetted up before it commences to deliver water to whatever sensing apparatus is connected to it and the transit time of that water from the funnel into the measuring system. It also reduces or eliminates evaporative losses, wind undercatch effects, and other issues. The arrival of the first drop of rain to strike an acoustic recorder can be recorded accurately with no delay, and the start and end times of rainfall can consequently be accurately identified. Furthermore, because more intense rain

generates acoustic energy that is directly linked to rainfall intensity, acoustic methods can, in principle, provide a direct means of recording intensity. Winder & Paulson [137] used a carefully designed tank of water, open to the rain and containing submerged microphones, to record sound energy from drops striking the water surface. They were able to estimate drop sizes and also interpret the acoustic signal in terms of intensity.

The acoustic detection of rainfall was adopted by Gray et al. [138] in the design of an automatic sequential rainwater sampler. In their device, rain struck a diaphragm whose area was 50 cm², mounted directly above a small loudspeaker. The voltage induced in the loudspeaker coil was used to drive a system for collecting a water sample and rotating a carousel of multiple collection bottles.

Potential problems with acoustic recording of rainfall are the influence of extraneous sound sources (wind, wildlife, vehicles) and the large data file sizes that result if high-resolution audio recording is used (easily amounting to Gb per day). However, it is possible to eliminate the problematic file size by recording the microphone or sensor signal much less frequently. Dunkerley [139] has demonstrated how this can be carried out with the logging of microphone signals at 10 s intervals.

It is worth noting that acoustic methods are also used in a manner similar to the use of radar. Bradley & Webb [140] developed an ultrasonic sodar (*sound detection and ranging*) system that sampled the incident drops in a volume of ~20 m³ and was quite sensitive at low rainfall intensities. A drop-counting rain gauge was used for validation. Likewise, Pang & Graßl [109] documented the development of a mini-sodar system capable of recording rainfall intensity.

2.7. Optical Rain Sensing Methods

Akin to the acoustic approaches just described, optical and image-based methods of detecting and recording rainfall—for example, detecting drops passing through a beam of light—can in principle offer faster response times than many types of gauges and other benefits. These include freedom from reliance on a collecting funnel and the associated uncertainties regarding rain start and end times, no losses to evaporation (Leeper & Kochendorfer [141]), and no lagged response of the kind associated with the progressive filling of a tipping bucket. Designs for optical rainfall measurement instruments have variously used infrared light frequencies (Salles et al. [99]), laser light (Stow et al. [142]), or visible light (Bradley et al. [143]).

Bradley et al. [143] described optical measurement of rainfall across a 2022 m path length between a light source, consisting of multiple 1000 W halide lamps and a detector (CCD camera). They accumulated images in 1 min time steps, and among these, 62% indicated rainfall rates of $\leq 1 \text{ mm h}^{-1}$ and 78% indicated rates of $\leq 5 \text{ mm h}^{-1}$. The approach clearly offers sufficiently high resolution to record low intensities (Figure 4). Results were compared with the data from four drop-counting rain gauges that collected accumulated rainfall depths every 15 s. Intensity was not directly recorded by light attenuation in a way akin to the measurement of optical turbidity but was calculated from the data on drop numbers as diameters and hence relied on the fitting of drop size distributions. The numerical inversion of light attenuation data in order to derive drop size and intensity data are not a straightforward process. Nor is the validation of intensity straightforward in the case of data collected over a >2 km path, since, for instance, wind and topographic effects often result in rainfall variation on km and shorter scales, as noted earlier. Further, it is possible that there may not be rain along the entire length of the light path.

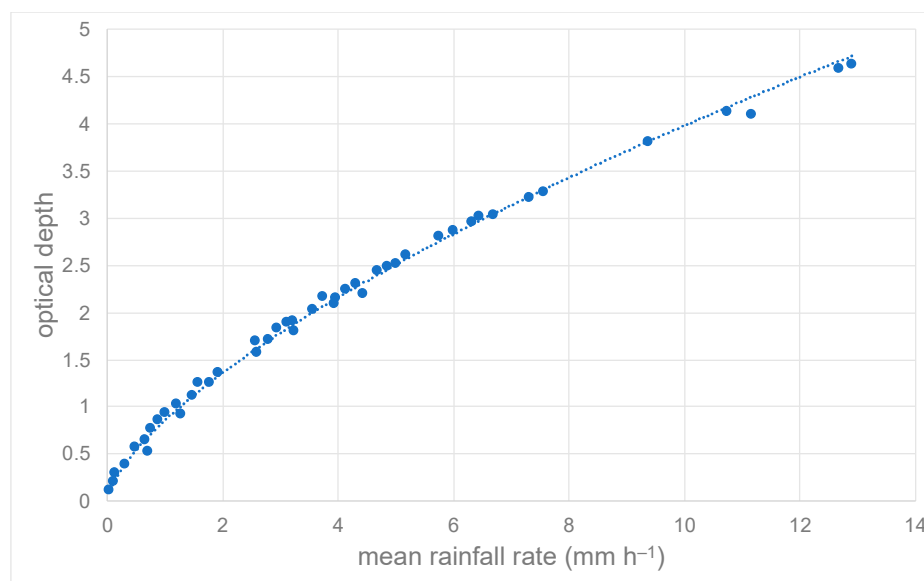


Figure 4. The relationship established by Bradley et al. (2000) [143] between optical depth measured along a 2220 m path and rainfall intensity was established using four drop-counting gauges. Illumination was visible light provided by halogen lamps. The regression model (dotted line) linking optical depth OD and rainfall rate (mm h^{-1}) has the equation $OD = 0.86 (\text{mm h}^{-1})^{0.667}$, for which $r^2 = 0.99$.

A contemporary source of drop imagery is the video data collected by cameras used for routine security and traffic monitoring. A significant benefit of a method using such cameras is the existence of huge numbers of potential rainfall observing sites, since security cameras in many countries now number in the millions, which greatly exceeds the number of meteorological rain recording stations. Allamano et al. [144] explored the use of camera imagery, estimating drop-fall speed from the streak length in images with known exposure times. Though requiring further development, they noted that their approach offered the potential of intensity data collected at 2 s intervals. How the image collection process would be triggered when rain fell (or discontinued when rain ended) is not clear. Jiang et al. [145] and Wang et al. ([146,147]) further developed camera-based methods. However, the use of the relatively low-quality imagery from surveillance cameras necessitates complex data processing, and the remaining uncertainties in estimates of intensity are large. As with other approaches to measuring intensity, the availability of independent, co-located measurements of intensity for validation purposes also remains a challenge.

Dong et al. [148] explored the recording of rainfall rate from video recordings, using the mean of 1000 successive frames to estimate the rain rate. They noted problems in handling intense rain when the ‘streaks’ marking the passage of individual drops were hard to identify separately. Photographic and video-based approaches to the recording of rainfall intensity seem to have the potential for considerable future refinement.

2.8. Thermal Rainfall Detection

Among the methods for recording intensity that have the potential to directly sense rainfall are those based on the evaporation of incident droplets and the tracking of the heating power required to achieve this. Battalino & Vonnegut [149] described such a device as capable of responding to rainfall intensities in the range of 0.3–350 mm h^{-1} . This consisted of a tin can wrapped in asbestos paper tape and wound with many turns of heating wire. The device, mounted with the cylindrical axis horizontal, had a projected collecting area of 79 cm^2 . Issues were that in very heavy rain, the available heating power might be insufficient to evaporate the water, or that splash, and drip may allow some incident drops to bypass the measurement process altogether, underestimating both the intensity and the aggregate rainfall accumulation. In heavy rain ($\sim 100 \text{ mm h}^{-1}$), the device

dissipated about 1 kW (Figure 5), which would prove difficult to supply in many locations. The interaction of a cylindrical sensor mounted horizontally with wind and its possible effects on catch efficiency have not been addressed.

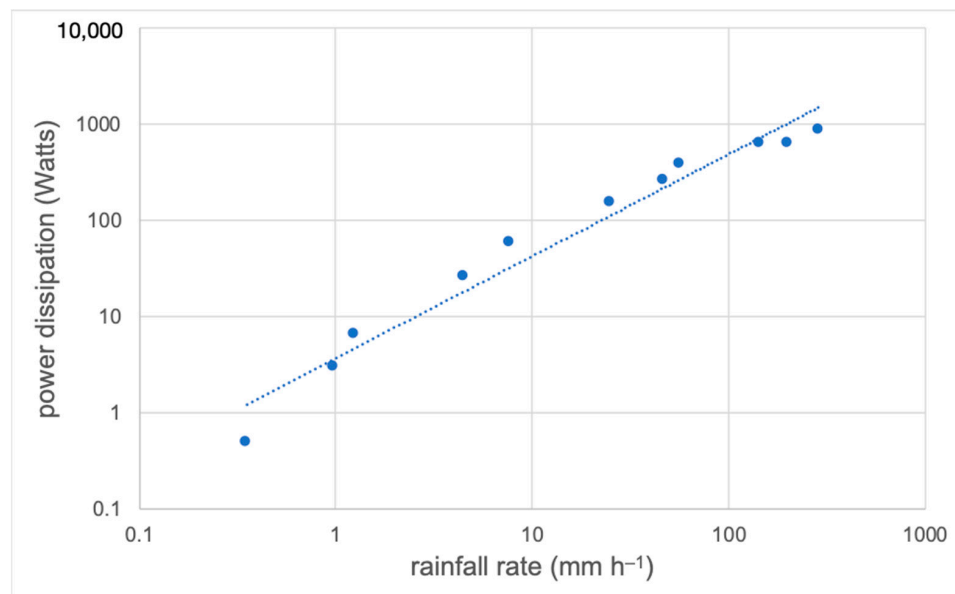


Figure 5. The relationship between heating power applied to evaporate rainwater and the incident rainfall intensity as measured by the rain intensity sensor of Battalino & Vonnegut (1978) [149]. Refer to the text for details. The regression model (dotted line) linking heating power (W) and rainfall rate (mm h⁻¹) has the equation $W = 3.71 (\text{mm h}^{-1})^{1.06}$ for which $r^2 = 0.96$ (adapted from Battalino & Vonnegut [149]).

Rasmussen et al. [150] devised a related device, which they called the “hotplate precipitation gauge.” This could record intensity in the range of 0.25–35 mm h⁻¹, with the capacity to record snowfall rates as well. The power to keep the ‘hotplates’ at a constant temperature was the signal used to track intensity. Data at 1 min temporal resolution were aggregated to 5 min data. This device has found most application in the measurement of snowfall (Cauteruccio et al. [151]), and a revised version of the device was described by Zelasko et al. [152].

It is interesting to note that thermal methods have also been used to detect the precise time of the commencement and cessation of rainfall, which conventional TBRGs cannot achieve. As an example, Raynor [153] devised and tested an electrical rainfall detector containing a heated, rotating cylinder with an array of electrodes that could signal the arrival of a single water drop. The device was located together with a TBRG having 0.01 inch (0.254 mm) sensitivity. Results revealed rainfall behaviour not recorded by the TBRG, including on one occasion the occurrence of 69 min of rainfall prior to the first TBRG tip event and 17 min of rain following the last TBRG tip event. Among the problems noted by Raynor [153] was the updraft of warm air from the heated drop detector. This apparently made the device incapable of recording dew or drizzle, and this issue might also arise with other thermal rain sensing devices, especially those using significant heating power.

2.9. Weighing Lysimeters

Lysimeters, consisting of an isolated monolith of soil and vegetation that is weighed by a load cell, have the potential advantage of being able to record the weight of incident rainfall without obstacles or wind-undercatch influences. However, in intense rainfall, it is possible for water to be lost as surface runoff. Nevertheless, the mass of a lysimeter can be recorded at frequent intervals, such as 10 min, therefore providing estimates of the rain rate (Kohfahl & Saaltink [154]). Studies that have compared the use of a lysimeter

with conventional rain gauges (e.g., Haselow et al. [155]) have quantified the kinds of discrepancies that can arise. Haselow et al. [155] employed a 4 tonne lysimeter weighed to 20 g and derived hourly rainfall data. Gebler et al. [156] used lysimeters with a 1 m² surface area in a grassland study in Germany. Multiple lysimeters were weighed at 5-s intervals with a resolution of 100 g, averaged to obtain 1 min data, and later aggregated to hourly rainfall data. The results were compared with measurements from a standard TBRG. For a test year (2012), the lysimeter recorded 16.4% more precipitation than the TBRG, which was mounted at 1 m above the ground. Various contributing factors were analysed, including the ability of the lysimeter to record dew.

In a study of dewfall made in Western Australia, Sudmeyer et al. [157] employed micro-lysimeters with a surface area of 0.075 m², which were briefly removed periodically for weighing to 0.1 g, the equivalent of 0.0013 mm of water over the area of the device. It proved possible to record dewfall rates of up to 0.032 mm h⁻¹. While this was a study of dewfall and not rain, this kind of device has the potential to sensitively record small amounts of low-intensity rainfall, which is challenging for many conventional methods of measurement, including TBRGs.

Morgan & Lourence [158] reported broad agreement between a grassed lysimeter 20 feet (~6.6 m) in diameter located in the Central Valley of California and several types of recording rain gauges located nearby. The resolution in rainfall depth was reported as 1/1000th inch (0.025 mm). Whether attempts were made to record intensity is not clear, and most of the 24 storms monitored lasted ~1 day.

Herbrich & Gerrke [159] emphasised the potential advantage of a lysimeter in having no surface differentiation from the surroundings and, given their typical size, a smaller impact from splash losses and wind effects on drop trajectories. They observed that even though it was possible to record lysimeter weight at frequent intervals, sources of vibration, such as wind, could result in apparent changes in weight that were not the result of precipitation. At a field site in Germany, they installed multiple lysimeters with surface areas of 1 m². Weight was recorded every 10 s and 1 min averages were logged. The resolution was 10 g, which corresponded to 0.01 mm of rainfall. As shown in Figure 2 of Herbrich & Gerrke [159], mass change correlated well with the rainfall rate (mm per 10 min). They reported that, as has often been found, the lysimeters recorded significantly higher accumulated rainfall depths, likely the result of the larger collecting area and diminished vulnerability to the wind undercatch that affects gauges. It is possible that, as a result, lysimeters might be capable of providing a more robust and literal estimate of the “ground truth” of rainfall arrival.

2.10. Other Electro-Mechanical Gauges

A wide diversity of rain gauge mechanisms has been proposed, and only examples are cited here. Förster et al. [160] reported the design and thorough evaluation of a piezo-electric rain gauge; a further design was presented by Henson et al. [161]. Figure 13 in Förster et al. [160] shows an excellent correspondence of piezo impact rainfall rates with those estimated by a co-located JWD. Erbakanov et al. [162] designed a gauge that counted drop impacts on a piezoelectric transducer. Xu & Zheng [163] developed a gauge in which the depth of water in a collecting chamber was measured using submerged ultrasonic transducers. Lan & Cao [164] proposed a gauge with a standard collecting funnel in which the water fell onto a cantilever beam, whose displacement was measured using a fibre Bragg grating. The device had a theoretical resolution of 15.4 µm of rainfall. This device appears to have been tested only under laboratory conditions.

There is a large group of devices that may be classified as electro-mechanical intensity gauges. Semplak [165] described a rapid-response gauge that used changing capacitance as water drained down a sloping trough to record rainfall intensity. Capacitance is challenging to use in this way, and the design was subsequently criticised and rejected by Fullerton & Raymond [166]. However, a successful capacitive design, again involving flow down a sloping channel, was described by Seibel [167]. Figure 8 in Seibel [167] shows the recording

of a rainfall event of 19 July 1971, in which the intensity lay between 0.1 and 1.0 mm for 72% of the rainfall duration. The device was capable of recording intensities up to 800 mm h^{-1} . Fullerton & Raymond [166] described and evaluated two other gauge designs, including the highly sensitive Workman gauge and their own sloping-channel design (the Raymond-Wilson gauge), which measured the electrical resistance across the channel between paired electrodes. In a field trial, these novel gauges were compared with the performance of a weighing gauge, a Jardí gauge, and a TBRG. All gauges had large (36 inch, approx. 914 mm) diameter funnel collectors, with the associated risks to system performance in low-intensity or intermittent rain already mentioned. None of these gauge designs appears to have found widespread application in routine rainfall observation.

2.11. Radiation-Based Methods

The final method for monitoring intensity uses nuclear methods. The few approaches to intensity measurement based on the use of radiation measurements include the use of changes in the atmospheric gamma-ray flux during rainfall, which permits the start and end times of rainfall to be determined, as well as measures of intensity (Zelinskiy et al. [168], Yakovleva [169]; Figure 6 below). Bottardi et al. [170] demonstrated that there was an increase in gamma activity related to an increase in the flux of ^{214}Pb during rainfall and were able to relate the two statistically. These methods appear to offer useful potential but require wider exploration and testing. Zelinskiy et al. [168] observed that gamma ray measurements, perhaps using dosimeters or gamma ray detectors, could be utilised for long-term monitoring of rainfall amounts and intensities.

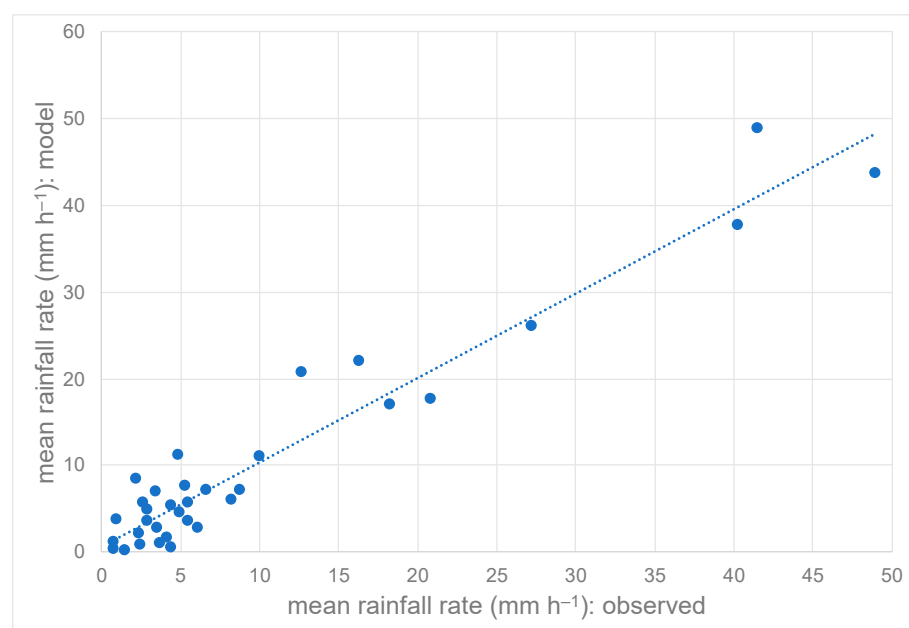


Figure 6. The relationship between the mean rainfall rate and the predicted rainfall rate is based on a model of gamma-ray dose. Rainfall was measured using a TBRG and a laser disdrometer. Data were collected at 1 min intervals. The regression model (dotted line) linking the modelled rainfall rate (RR_m) and the observed rainfall rate (RR_o) has the equation $RR_m = 0.97(RR_o) + 0.71$, for which $r^2 = 0.93$ (adapted from Zelinskiy et al. [168]).

3. Methods Suitable for Wide-Area Rainfall Rate Measurement

Intensity recording methods discussed in Sections 2.1–2.11 above yield data at a single location on the ground. However, for catchment-based studies, data distributed spatially is needed, particularly for localised convective storms. One approach to gathering such data are to install multiple gauges; however, it is also possible to adopt measurement methods

that have the capability of recording area-wide intensity data. Three such approaches are considered next.

3.1. Radar-Based Approaches

There are many approaches to the recording of rainfall arrivals using some form of radar detection of droplets. Radar systems may be operated from the ground or from satellite platforms. Essentially, they detect the strength of the backscattered part of the radar signal, Z , which increases with the volume occupied by droplets in the area scanned, essentially linked to the rainfall rate R . Thus, interpretation depends on establishing, for different kinds of rainfall (stratiform, convective, etc.), the Z - R relationship (e.g., Kirsch et al. [171]). There are many different radar configurations (frequencies used, beam dimensions and polarization, scan frequencies, and other parameters), and it is neither possible nor appropriate to cover these in this review. Reviews of radar methods for precipitation measurement include Wilson & Brandes [172], Atlas [173], Sauvageot [174], Krajewski & Smith [175], Nanding & Rico-Ramirez [176], Borga et al. [177] and for satellite-based methods, the two volumes edited by Levizzani et al. [178]. Radar applications include large systems capable of scanning hundreds of km^2 , small systems such as the micro-rain radar (MRR), and relatively new devices, including the Lufft WS100 low-power radar sensor that is smaller than many conventional rain gauges (see <https://www.lufft.com>, accessed on 2 April 2023). A recent test of this device by Vokoun & Moravec [179] found that the device reported rainfall that was much larger than the result from conventional gauges. Some implementations of radar systems provide a disdrometer capability. Prodi et al. [180] for instance, explored the Pludix X-band doppler disdrometer, which senses over an area of a few square metres above the radome.

Mansheim et al. [181] reported the performance of a microwave rain radar. This was in fact an adaptation of a radar unit intended for the measurement of farm vehicle ground speed but oriented upward from the ground instead of downwards as intended for use on agricultural machinery. Using the Marshall-Palmer model of the rain DSD, this configuration yielded a signal proportion to the mean speed of droplets within the field of view, independent of the volume being sensed (see this relationship plotted in Figure 4 in Mansheim et al. [181]), and this was then related to the rain rate in mm h^{-1} .

Chang et al. [115] describe the conversion of MRR estimated drop diameters to fall speed using the Gunn & Kinzer [89] relationship. They deployed 16 instruments, including five vertically pointing MRRs, within an area of 400 m^2 . Measurements over several weeks were collected in order to compare the representation of DSDs. The MRR showed higher concentrations of small drops ($<1.0 \text{ mm}$) than did 2DVD or JWD, and also more large drops ($>5.2 \text{ mm}$). Chang et al. [115] attributed this to the larger sampling volume of the MRR and the resulting better representation of the less frequent arrival of large drops and of the more difficult detection of small drops. The JWD was found to be the least accurate, and the MRR had the lowest uncertainty owing to its larger sample volume and accurate Doppler measurement principle.

Though weather radar systems are widely used to record precipitation and now cover the range from wide-area measurement to virtual at-a-point measurement, the approach is an indirect one, and calibration is required using existing forms of rain gauge, commonly the TBRG. The great advantage offered by radar-based methods, however, is their ability to provide area-wide data with high temporal resolution, such that the progression of storm cells across the landscape can be monitored. Quantitative rainfall measurements are still problematic and affected by many influences other than rainfall itself, including ground clutter (Krajewski et al. [182]); performance in heavy rainfall remains relatively poor (Pastorek et al. [183]).

3.2. Microwave Attenuation (Cellular Phone Links, Satellite Links, etc.)

Path-based approaches to the estimation of rainfall rates have been widely explored. These include attenuation on commercial microwave links (CMLs) such as those that form cellular telephone networks (Roversi et al. [184]), as well as links that carry data to and from satellites. Lian et al. [185] provided a recent review. Essentially, the more intense the rainfall along the microwave path from transmitter to receiver (mm h^{-1}), the greater the signal attenuation (dB). Figure 2 in Lian et al. [185] illustrates this relationship with data from two days with intermittent rainfall. The potential utility of these CML-based methods is that there is a large network of devices already installed, and the network is particularly dense in urban areas where rapidly acquired rainfall data could be of great value in the prediction of flash flooding. A well-known issue is that attenuation additional to that caused by the rainfall along the link path can arise from water adhering to the antennas; this can result in the overestimation of rainfall rates. However, devices and procedures to correct for this are being developed. Nebuloni et al. [186] conducted a field test in Italy, in which CML data were compared with rain gauge and disdrometer data. Their results suggest that the use of CML data are successful in detecting the occurrence of rainfall, but less so in quantifying rainfall amounts or intensities.

Giannetti et al. [187] and Giannetti & Reggiannini [188] explored the determination of rainfall rate from the attenuation of satellite downlink data, which involves transmission frequencies in the range $\sim 10\text{--}40$ GHz. They reviewed the difficulties that arise from rain occupying only a part of the beam path, resolving rain start and end times, and other factors that affect the method.

Additional explorations of the use of CML attenuation data in the recording of rainfall include Kumah et al. [189], Pudashine et al. [190], and Zheng et al. [191]. Zheng et al. [191] focus particularly on the capability of CML data to provide good area rainfall data in urban areas. Recent studies in general confirm that with careful processing, CML data can provide distributed rainfall data that can be applied with confidence to studies of urban hydrology and flooding problems (Liu et al. [192], Pastorek et al. [183]).

3.3. Seismic Methods for Recording Rainfall

The impact of rainfall on the ground creates sound waves that travel through solid earth materials and can be recorded by seismic geophones. Indeed, rain is a source of environmental seismic noise that is known to have the potential to interfere with the intended purpose of seismic work in areas such as geological investigations (Dean [193]).

Bakker et al. [194] explored the seismic recording of rainfall in a catchment in France, using Parsivel disdrometer data to provide 1 min rainfall data. They found that most seismic energy recorded at a geophone arrives from a radial distance of up to ~ 25 m from the sensor. Because up to 90% of the seismic power was found to arise from drops of >3 mm diameter, Bakker et al. [194] suggest that seismic monitoring is best suited to the study of intense rainfall, during which large drops arrive more frequently.

Diaz et al. [195] showed that at frequencies above 40 Hz, most seismic signals at their field site in Spain were generated by rainfall. This then suggested that frequencies above this threshold could be used to monitor rainfall. They were able to collect seismic data at 6 min intervals and monitor the passage of rainfall cells across the landscape using a spatially distributed network of seismic stations.

Like microwave attenuation and radar methods, seismic monitoring of rainfall has the potential advantage of providing spatial coverage and sensing areas much larger than those of gauges, disdrometers, or other point-located devices. However, further exploration of these methods will be necessary to explore their suitability on geological substrates of varying properties and within the range of vegetation canopies, which may alter drop speeds and kinetic energy, as well as drop sizes.

4. Miscellaneous Approaches to Rainfall Recording

The small drop-collecting area of several types of rain gauges was mentioned previously. Grimaldi et al. [196] explored radically increasing the collecting area of a rain gauge to 100 m². Unsurprisingly, this resulted in challenges related to evaporative losses and the system's response time to rapid intensity changes related to the long flow paths of water draining into the measuring system.

Novel approaches to rainfall recording continue to be devised. Moving vehicles provide a potentially large, distributed network of observing sites. This possibility was explored in numerical simulations (Haberlandt et al. [197], Calafate et al. [198]) and in rainfall simulation experiments in the laboratory (Rabiei et al. [199]). The approach used by Rabiei et al. [199] was to relate the windscreen wiper speed, adjusted either by a human 'driver' or automatically based on a sensor, to the rainfall intensity. Issues relate to the effect of windscreen angle, vehicle speed, the nature of the rain (intensity, drop size distribution), and, in the case of manual wiper speed adjustment, the judgement of the driver as to when visibility has become poor. This field was extended by Kim et al. [200], again using rainfall simulation. All studies managed to establish a relationship between rainfall intensity and wiper speed, but with quite wide uncertainty. As Kim et al. [200] put it, however, somewhat uncertain rainfall data from a very large number of vehicle observations might have value comparable to precise data from a much smaller number of meteorological stations; they point out that even data from 1% of the then ~20 million registered vehicles in Korea would provide a very large data set, distributed spatially. Of course, there would be a need for data networks to be established in order to gather these data in real time for them to be of use for flash-flood prediction or related purposes. Currently, there appear to have been no field trials using many vehicles under real driving conditions.

The microphone function of smartphones has been explored as an alternative source of multitudinous observations across the landscape. Gaucherel & Grimaldi [201] developed the 'Pluviophone'; Guo et al. [202] developed 'Chaac', a system consisting of an umbrella and an attached smartphone for acoustic data collection. They analysed 1 s long audio segments and 10 s long 'clips' from which to judge intensity.

The use of motor vehicles and mobile phones for the collection of rainfall data are instances of the growing adoption of citizen science or 'crowdsourced' data collection at multiple sites (for synchronised, spatial coverage). For instance, Mapiam et al. [203] concluded that citizen observations of rainfall could improve the correction of the bias in radar measurements of rainfall in the Tubma Basin, Thailand.

Rainfall over the oceans continues to warrant research targeting suitable methods of measurement, including devices that can be attached to tethered buoys. Förster et al. [160] described a piezoelectric gauge, with a spherical impact sensor, designed for use on buoys. Chai et al. [204] described a self-siphoning gauge with capacitive measurement of rainfall that was designed for use in the marine environment. Lu et al. [205] present a method for interpreting rainfall intensity at sea from ship-borne X-band radar. The use of acoustic methods for recording rain at sea has become an established method and has already been described.

5. Discussion and Conclusions

The foregoing discussion makes it clear that the widely-used TBRG is not particularly well-suited to the recording of rainfall duration, intermittency, or intensity, particularly in rain of low intensity or where there are multiple rainfall events per rain day. Nevertheless, it was shown that the TBRG is widely used to provide the "ground truth" against which to evaluate many other methods of measurement, including disdrometers, radar, and others. A better 'standard' is therefore really needed. Most rain gauges, whether weighing, drop-forming, tipping buckets, etc., rely on a collecting funnel to gather sufficient drops of rainfall to be measurable. However, such a collecting funnel takes time to wet up sufficiently to drain into the measurement apparatus below; initial drops may simply come to rest on the funnel surfaces (Camuffo et al. [206]), and droplets may remain there when rain ceases, and

never be tallied at all. (To overcome this problem, Woo & Steer [207], working in the Arctic, recorded very small amounts of precipitation by re-weighing blotters that were exposed to rain, avoiding any wetting-up thresholds.) In relation to droplet adhesion and evaporative losses from collecting funnels, we can note that the surface area of a 300 mm diameter rain gauge funnel, with the conical sides sloping inward at 45° , is almost twice the area of the funnel opening. The inclined surface that can support evaporative losses is $\sim 1346 \text{ cm}^2$, and not the $\sim 700 \text{ cm}^2$ of the funnel opening.

Multiple studies have shown that under windy conditions, there are various micro-physical and related effects on the drop size distribution and on fall speeds and trajectories. The magnitude of the effects may vary with the type of rainfall (stratiform or convective) or during a single event as the nature of rainfall and/or the windspeed vary. Small drops have been found to commonly exhibit super-terminal fall speeds, and large drops, sub-terminal speeds (Niu et al. [91], Bringi et al. [92], Chatterjee et al. [93]). Corrections can be made if the wind speed and drop trajectories are also measured sufficiently often during rainfall, however, in the absence of such corrections, errors in mean rainfall rate as recorded by a Parsivel disdrometer through 5 min, in wind at 14 m s^{-1} , can amount to $>-22\%$ (Lin et al. [107]). Nevertheless, these studies show that a reliance on relationships between fall speed and diameter established in stagnant air (such as the widely-used data of Gunn & Kinzer [89]) cannot be reliably applied to the conversion of disdrometer data collected under windy conditions, such as that from a JWD, into rainfall rate data. In this context, it is worth mentioning the novel rain gauge design presented by Stewart et al. [208], in which an ordinary catching gauge was attached to a long, flexible rod that was free to oscillate in the wind. The frequency of oscillation could be linked to the mass of rainwater contained in the attached gauge, mounted near the top of the flexible rod, and the device could also be used to monitor wind speed. This device would require elaboration and refinement for field application, however, having, for instance, no means to self-empty the volume of rainwater held in the gauge.

Rainfall intensity is rarely constant and indeed typically fluctuates continuously from moment to moment, to judge from the available data collected with a timing resolution of seconds or indeed with the essentially instantaneous data provided by acoustic recordings in which sound pressure is logged at 44.1 kHz. The rates at which intensity changes during rainfall have rarely been quantified; however, several studies show that they can reach $\sim 800 \text{ mm h}^{-1} \text{ min}^{-1}$ (Table 5 in Dunkerley [209]) when estimated from field data having very high temporal resolution. Examples of such high-resolution data can be found in Figure 12 of Stow & Dirks [78] which presents data from Norfolk Island, including a very rapid increase from 50 to 454 mm h^{-1} at a rapid-response rain gauge and some equally rapid declines in intensity. Figure 7 presents some acoustic data recorded during rainfall in the Australian wet tropics, with the microphone signal recorded only every 10 s to reduce the volume of data collected. Rapid jumps in intensity are nevertheless clearly evident in this recording.

Intermittency is problematic to quantify accurately in the absence of a true intensity gauge because, as noted above, rain may begin and end from moment to moment and often does so many times during an hour. Examples of the intermittency of rain, collected with devices providing high temporal resolution, can be seen in many time-series plots (Figure 1 in Bradley et al. [143], Figure 6 in Henson et al. [161], and Figures 11–13 in Förster et al. [160]).

Acoustic gauges are capable of recording unambiguously the moment of arrival of the first few drops of rain and likewise of indicating when drop arrival has ceased. Commercial sensors operating on the acoustic detection of drop impacts are available. An example is the “RHD” (rain and hail detector) rain sensor from Sommer Messtechnik, Austria (<https://www.sommer.at/en/>, accessed on 2 April 2023), which employs a polished stainless-steel hemisphere 160 mm in diameter (surface area 402 cm^2) (Figure 8). This device has no moving parts. In this sense, acoustic gauges offer a path to the design of true intensity gauges, which respond to changes in rainfall arrival from moment to moment. Likewise,

evaporation after droplet arrival at the gauge does not affect the signal of raindrop arrival that has already been recorded in an acoustic gauge. The response of electronic leaf-wetness gauges to the arrival of the first few drops of water is similarly rapid, and dew gauges can likewise record the first condensed droplet (e.g., Camuffo et al. [210]). The WS100 radar gauge is also said to be able to respond to first-drop arrivals at the sensor.

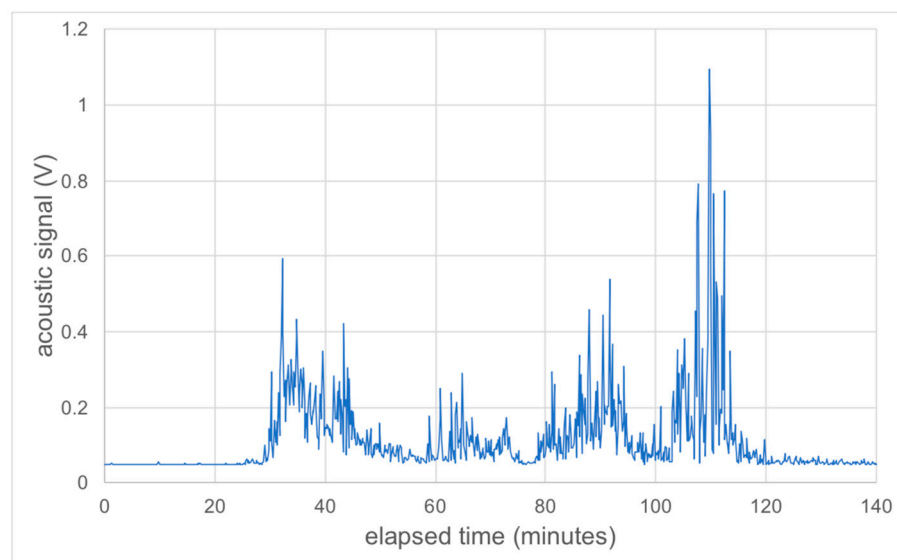


Figure 7. An acoustic recording of rainfall was collected using the acoustic sensor at Dunkerley (2020) [136]. The output signal from a microphone was logged at 10 s intervals. The rainfall event shown had a duration of 1 h 46 min and delivered 5.6 mm of rain at an average intensity of 3.2 mm h^{-1} . The plotted time series of acoustic signals contains 924 voltage readings and reveals the rapid fluctuations in intensity that could not be captured in the record of just 28 bucket tip events derived from the TBRG. The highest intensity peak (1.1 V) visible between 110 and 120 min of elapsed time is equivalent to $\sim 55 \text{ mm h}^{-1}$ (source: field data by the author).



Figure 8. The “RHD” acoustic rainfall and hail sensor of Sommer Messtechnik (Austria). The sensor, which has no moving parts, is a hemisphere of stainless steel with a diameter of 160 mm (see <https://www.sommer.at/en/>, accessed on 2 April 2023).

5.1. Point Rainfall Intensity vs. Areal Rainfall Intensity Data

We have seen that there are distinct challenges in attempts to measure rainfall intensity at a single location versus spatially, such as the area-wide mean rate at any instant over a drainage basin. Approaches to area rainfall intensity measurement using rain radars of various kinds, which can provide data over hundreds of km², require validation against “ground truth” data. This is problematic since there are no existing methods to achieve area intensity measurements at the ground apart from dense networks of rain gauges (e.g., see Kirchengast et al. [211]). Even short-path commercial microwave link (CML) attenuation measurements over a few km cannot readily be compared numerically with point-based data, such as disdrometer data, since again, there is no ready means of making the data accordant. Lack of correspondence can arise because even a dense network of conventional rain gauges may not sample precisely the same area-wide field of rainfall that is scanned by radar or along a CML link. Additionally, both radar and CML network data relate to elevations above the ground (Overeem et al. [212] while gauges are typically located at heights <0.5 m above the surface. During the fall trajectory below the radar scan, droplets can drift laterally on the wind and change diameter owing to evaporation. These phenomena were examined by Dai et al. [213] using data from a site in the UK. They showed that the downwind drift of raindrops could amount to several kilometres for large drops (5.0 mm diameter) and up to 14 km for small drops (0.2 mm). They also reported that drop diameter can decline during passage to the ground as a result of evaporation, reducing the intensity below that suggested by radar scans made before drops fell to ground level.

Point-based rainfall data are desirable for some applications, such as data on soil loss from a small plot or field, while areal data are desirable at larger scales, such as when attempting to validate models of streamflow from catchment areas, especially those with short response times (post-fire landscapes, urban areas). Remotely sensed rainfall data have the capacity to provide spatial rainfall data, such as the 0.01° spatially gridded data explored by Hirose & Okada [4], and multiple point-gauges can be deployed to achieve good spatial resolution of rainfall fields. Perhaps the best-known example is the network of 88 gauges in the 149 km² Walnut Gulch catchment in SE Arizona, USA (<https://data.nal.usda.gov/dataset/walnut-gulch-experimental-watershed-arizona-precipitation>, accessed on 2 April 2023). Other dense networks of gauges have been established for particular research purposes, such as the 96 rapid-response gauges set out over an area of <180 km² in New Jersey, USA (Freeny & Gabbe [214]), all providing remotely-read data at 1 min temporal resolution. Multi-gauge installations of this kind are uncommon and are unlikely ever to be sufficiently widespread as to provide data on rainfall intensities at all the locations globally where this would be advantageous. Maier et al. [215] tabulated details of multiple studies employing dense gauge networks. Such studies have been applied extensively in urban areas for purposes of exploring rainfall data requirements for flood studies (Yoon & Lee [216]), but also in rural and mountainous catchments (Villarini et al. [217], Sucozhañay & Céleri [218]).

5.2. What Can We Say about Likely Future Developments in the Measurement of Intensity?

Sivapalan & Blöschl [219] outline a theory of what drives innovation in hydrological methods and instrumentation. They point to cycles of euphoria followed by periods of disenchantment that drive change and innovation.

Has an optimum method for the recording of rainfall intensity been found? Should we be euphoric or disenchanted? The answer depends on various issues, including establishment costs in relation to the available budget, the capacity to maintain instruments or instrument networks, the spatial extent of the area over which rainfall intensity data are sought, and many others. Power consumption, robustness, and suitability for long-term deployment are also important parameters.

Undoubtedly, the development of new approaches to the recording of rainfall and rainfall intensity will continue. There have been trials of community-sourced (‘crowd-sourced’) data from ‘citizen scientists’ (Niu et al. [220]). These include incorporating data

from personal weather stations and other ‘opportunistic’ sensors (Graf et al. [221]). The use of opportunistic sensors has also been extended to the processing of vision from pre-existing camera networks (Jiang et al. [145]), and in confirming the occurrence or absence of rainfall by using dashcam vision showing the activity of vehicle windscreen wipers (Bartos et al. [222]). For urban sites, Yin et al. [223] have trialled a method based on the training of a convolutional neural network. They analysed images collected by smart phones but needed to make allowance for the very brief exposure time ($\sim 1/200$ s), which could not be directly validated against even 1 min of data from conventional rain gauge data. If this approach can be developed further, then it may be possible for community-sourced images to be pooled to provide ‘snapshots’ of rainfall behaviour across extensive regions, based on many more ground observations than can be achieved with existing networks of meteorological stations.

There are thus at least two frontiers in rainfall intensity research:

- (1) the challenge of measuring at a point with high temporal resolution and of finding a method prone to as few drawbacks or sources of bias as possible;
- (2) the related challenge of measuring spatially-distributed rainfall characteristics, for instance across an experimental catchment or an urban area prone to flash flooding.

The two are related, not least because any area measurement method requires validation at locations within the area covered. Microwave transmission paths may be tens of km in length, and across the network of towers, they may cover very large areas. The same issue applies to sources of remotely sensed rainfall data, such as that from TRMM and its successor, the GRMP. For instance, Hirose & Okada [4] developed gridded rainfall TRMM data at 0.01° resolution. Their data showed considerable divergence among data gridded at 0.1° and 0.01° , including in the distribution of rainfall across rugged topography. A notable instance was their finding that mean daily rainfall is higher (reaching an average of 28.9 mm d^{-1}) at a location 5 km east of Cherrapunji, in the Meghalaya Hills, India, than at Cherrapunji itself, one of the wettest locations on Earth. This result lacks ground validation and does not provide the high temporal resolution needed to explore storm types, the fluctuating intensities during rain, and so on. It has to be kept in mind that a successful validation of an approach to recording rainfall intensity in a particular test location does not ensure that the method will be successful in other locations. Many areas across which rainfall data would ideally be available are too rugged or too remote to be instrumented with ground-based devices. However, these may be just the locations where rugged or complex topography may induce wind-related effects that were not included at the site of the prior successful validation. It seems inevitable, therefore, that for some spatially dispersed data, for instance, from satellite-based remote sensing, there may be local, terrain-related biases that it may never be possible to document with the exactness that might be desired.

The diversity in the temporal resolution of rainfall data remains a problem. Time-averaged intensity (that is, rainfall rate) is widely known to decline with longer aggregation times. Hourly data remain the most commonly available, but say little about actual intensities and nothing at all about sub-hourly intermittency. On the other hand, some of the methods reviewed here offer resolutions of 10 s or finer. While some aggregated data may satisfy the requirements of descriptive climatology, data that are closer to true intensities are required for many of the discipline areas mentioned earlier (erosion and related investigations, flash flood studies, and to document and quantify intensity changes that may be associated with climate change and variability). Therefore, it can be concluded that many tasks remain to be addressed in ongoing research targeting the fine-scale arrival characteristics of rainfall and that improved methods and protocols for the measurement of rainfall intensity are worth pursuing.

Funding: This research received no external funding.

Data Availability Statement: No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Conflicts of Interest: The author declares no conflict of interest.

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